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# Integrability and Tree-Level OPE Coefficients in ABJM: Combinatorics through Functional Equations

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Abstract:

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## 1 To do

**P: add more later**

1. How are the fields charged under $SU(4)$? Identify them explicitly with basis states in fund and anti-fund of $SU(4)$ (it is in the Klose review)
2. Untwisted + twisted QQ solver (no $\theta$ necessary)
3. Build transfer matrices explicitly for $L = 2, 4$ and write code to compute spin chain overlaps corresponding to Wick contractions.
4. What is the "Yangian" symmetry (in the sense of asymptotic 4pt paper) of the structure constant (in N=4 the structure constant vanishes unless the excited state is LR symmetric)
5. Construct det of Qs which makes that symmetry manifest.

## 2 Introduction

Integrability has emerged as a powerful tool in studying the AdS/CFT correspondence.

Hexagons don't work for ABJM for some reason (why?). There are no systematic methods for structure constants in high-rank sectors (some partial progress in $\mathfrak{su}(2)\times\mathfrak{su}(2)$).

In this paper we present simple determinant formulas for structure constants in the full $\mathfrak{su}(4)$ sector.

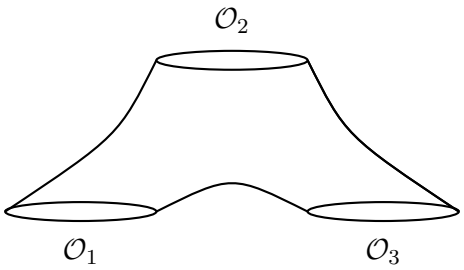



**Figure 1**: Caption

## 3 Reminder of ABJM

ABJM theory is a superconformal Chern-Simons theory coupled to matter. Local operators are constructed from traces of the spinors $\psi_a, \psi^a$ and the scalars $Y_a^\dagger, Y^a$. In this paper, we will only concern ourselves with the scalars, the so-called $\mathfrak{su}(4)$ sector of the theory. This is the sector made out of operators of the form

$$\mathcal{O} = C^{a_1 a_2, \ldots a_L}_{b_1 b_2 \ldots b_L} \operatorname{tr} Y^\dagger_{a_1} Y^{b_1} \ldots Y^\dagger_{a_L} Y^{b_L} \,. \tag{3.1}$$ {?}

In particular, we are interested in computing 3pt functions among $\mathcal{O}$. 3pt functions are heavily constrained by conformal symmetry and must take the form

$$\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_3(x_3)\rangle = \frac{1}{N}\frac{C_{123}}{|x_{12}|^{2(\Delta_1+\Delta_2-\Delta_3)}|x_{13}|^{2(\Delta_1+\Delta_3-\Delta_2)}|x_{23}|^{2(\Delta_2+\Delta_3-\Delta_1)}} \tag{3.2}$$ {?}

where we have stripped a factor of $N$ from the structure constant for convenience. In this paper, we will consider the special case when $\mathcal{O}_2, \mathcal{O}_3$ are BPS operators. To be precise, we will consider the configuration

$$\mathcal{O}_2 = \operatorname{tr}\left(Y_1^\dagger Y^4\right)^L \,, \quad \mathcal{O}_3 = \operatorname{tr}\left(\tilde{Y}_1^\dagger \tilde{Y}^4\right)^L = \operatorname{tr}\left(\left(Y_1^\dagger - Y_4^\dagger\right)\left(Y^1 + Y^4\right)\right)^L \,, \tag{3.3}$$ {?}

while the excited operator takes the form

$$\mathcal{O}_1 = \operatorname{tr}\underbrace{\overbrace{X_1 X_1^\dagger X_2 X_2^\dagger}^{2M} Y^1 Y_4^\dagger \ldots Y^1 Y_4^\dagger}_{2L}, \tag{3.4}$$ {?}

i.e we view it as an excitation over the vacuum

$$\mathcal{O} = \operatorname{tr}\left(Y^1 Y_4^\dagger\right)^L \tag{3.5}$$ {?}

This maps the problem to the following computation

$$\frac{C^{\bullet\circ\circ}_{123}}{C^{\circ\circ\circ}_{123}} = \mathcal{N}_{123}\left(\left\langle Y_1^\dagger Y^4\right|^{L-M} \otimes \left\langle \tilde{Y}_1^\dagger \tilde{Y}^4\right|^{\otimes M}\right)|\Psi\rangle \tag{3.6}$$ {?}

## 4 Integrable Alternating Spin Chains

Let us recall the standard structure of the integrable alternating $\mathfrak{su}(4)$ spin chain.

We h

$$Y^a \mapsto \mathbf{e}^a\,, \quad Y_a^\dagger \mapsto \mathbf{e}_a \tag{4.1}$$ {?}

where $\langle \mathbf{e}_a, \mathbf{e}^b \rangle = \delta_a^b$

We will in this work need the Lax-matrices. Let us introduce matrices $\mathbf{e}_a{}^b$, the Lax-matrices takes the form

$$\mathcal{L}_{\mathbf{4}\mathbf{4}} = (u - \frac{i}{2}) + \mathbb{P}\,, \quad \mathcal{L} \tag{4.2}$$ {?}

Using these Lax-matrices we can build 3 (twisted) monodromy matrices,

$$\mathcal{T}_a^G = \mathcal{L}_{a\mathbf{4}}(u-\theta_1)\mathcal{L}_{a\bar{\mathbf{4}}}(u-\bar{\theta}_1)\dots\mathcal{L}_{a\mathbf{4}}\mathcal{L}_{a\bar{\mathbf{4}}}G \tag{4.3}$$ {?}

and we set

$$Q_\theta = \prod_{\alpha=1}^{L}(u-\theta_\alpha)\,, \quad Q_{\bar{\theta}} = \prod_{\alpha=1}^{L}\left(u-\bar{\theta}_\alpha\right)\,, \tag{4.4}$$ {?}

The spectrum of these transfer-matrices is given as

$$\mathbb{T}_{\mathbf{4}} = Q_\theta Q_{\bar{\theta}}^{[-2]}\frac{q_1^{[3]}}{q_1^+} + Q_\theta^{[2]}Q_{\bar{\theta}}^{[-2]}\frac{q_{12}^{[2]}}{q_{12}}\frac{q_1^-}{q_1^+} + Q_\theta^{[2]}Q_{\bar{\theta}}^{[-2]}\frac{q_{123}^+}{q_{123}^-}\frac{q_{12}^{[-2]}}{q_{12}} + Q_\theta^{[2]}Q_{\bar{\theta}}\frac{q_{123}^{[-3]}}{q_{123}^-} \tag{4.5}$$ {?}

Absence of poles gives the following set of Bethe equations

$$\frac{q_1^{[2]}}{q_1^{[-2]}}\frac{q_{12}^-}{q_{12}^+} = -\frac{Q_\theta^+}{Q_\theta^-}\,, \quad \frac{q_{12}^{[2]}}{q_{12}^{[-2]}}\frac{q_1^-}{q_1^+}\frac{q_{123}^-}{q_{123}^+} = -1\,, \quad \frac{q_{123}^{[2]}}{q_{123}^{[-2]}}\frac{q_{12}^-}{q_{12}^+} = -\frac{Q_{\bar{\theta}}^+}{Q_{\bar{\theta}}^-} \tag{4.6}$$ {?}

This gives the following set of transfer matrices

$$\tau_{\mathbf{4}} = I_{2L}u^{2L} + \sum_{\beta=0}\,, \quad \tau_{\mathbf{6}} = I_{\mathbf{6},2L} + \sum\,, \quad D \tag{4.7}$$ {?}

giving us a total of 6

### 4.1 Integrable Boundary States

Let us introduce the concept of an integrable boundary state, it is a state $\langle W|$ such that

$$\langle W|\,(\hat{\tau}_{\mathbf{4}} - \hat{\tau}_{\bar{\mathbf{4}}}) = 0\,. \tag{4.8}$$ {?}

Let us introduce

$$\langle W_a| = (\mathbf{e}_1 + a\,\mathbf{e}_4)\otimes(\mathbf{e}^4 - a\,\mathbf{e}^1) \tag{4.9}$$ {?}

### 4.2 Derivation of inner product

Let us now derive an inner product for $Q$-functions. We consider the requirement that $\tau_{\mathbf{4}} = \tau_{\bar{\mathbf{4}}}$

we will use the language of $Q$-operators. We recall that in an $\mathfrak{su}(4)$ Q-system we have $Q_a, Q_{ab}, Q_{abc}$, which takes the form

$$Q_a \simeq \lambda_a^{-iu} q \tag{4.10}$$ {?}

where $q_a$ are polynomials in $u$. Importantly, they satisfy the following Baxter equation

$$q^{[4]} - \tau_{\mathbf{4}}^{+} q^{[2]} + \tau_{\mathbf{6}}\, q - \tau_{\bar{\mathbf{4}}}\, q^{[-2]} + q^{[-4]} = 0 \tag{4.11}$$ {?}

where $\tau_4^+, \tau_6, \tau_{\mathbf{4}}$ are all polynomials.

### 4.3 SoV in the Alternating Spin Chain

The SoV program starts from an inner form between Q-functions, the case of alternating spin chains was considered in [1]. We will follow the main notation introduced in that paper, we have a Baxter equation

$$\mathcal{O} = \frac{1}{Q_{\bar{\theta}}^{+} Q_{\bar{\theta}}^{-}} \underbrace{\left( Q_{\bar{\theta}}^{-} D^4 Q_{\theta}^{+} - \tau_{\mathbf{4}}^{+} D^2 + \tau_{\mathbf{6}} - \tau_{\bar{\mathbf{4}}}^{-} D^{-2} + Q_{\bar{\theta}}^{+} D^{-4} Q_{\theta}^{-} \right)}_{\mathcal{O}_R} \frac{1}{Q_{\theta}^{+} Q_{\theta}^{-}} \tag{4.12}$$ {?}

and hence we introduce

$$\mathbf{Q}_i = \frac{Q_i}{Q_{\theta}^{+} Q_{\theta}^{-}}\,, \quad \mathbf{Q}^i = \frac{Q^i}{Q_{\bar{\theta}}^{+} Q_{\bar{\theta}}^{-}}\,. \tag{4.13}$$ {?}

so that

$$\mathcal{O}\, \mathbf{Q}_i = 0\,, \quad D \tag{4.14}$$ {?}

We introduce an inner product

$$\langle f g \rangle = \oint du\, \rho\, f g \tag{4.15}$$ {?}

We need to build an inner product so that

$$\langle f \mathcal{O} g \rangle = \langle f \mathcal{O} g \rangle \tag{4.16}$$ {?}

There are only a few linearly independent functions, we pick

$$\rho_\alpha = \frac{1}{\mathcal{R}_\alpha} \prod_{\beta \neq \alpha}^{L} \left(1 + e^{2\pi(u - \theta_\beta)}\right) \prod_{\beta}^{L} \left(1 + e^{2\pi(u - \bar{\theta}_\beta)}\right) \tag{4.17}$$ {?}

**The Baxter Equation** A main player in the integrability

## 5 Explicit OPE Coefficients

## A Baxter for $\mathfrak{su}(4)$

In this section, we present the Baxter equation for a rational spin chain with Hilbert space $\mathcal{H} = \mathbf{4}^{\otimes L_{\mathbf{4}}} \otimes \mathbf{6}^{\otimes L_{\mathbf{6}}} \otimes \bar{\mathbf{4}}^{L_{\bar{\mathbf{4}}}}$ .

To derive the Baxter equation, we start from the following Q-system

$$Q_{ab} = \mathrm{Wr}\left(Q_a, Q_b\right) , \quad Q_{abc} = \mathrm{Wr}(Q_a, Q_b, Q_c) , . \tag{A.1}$$ {?}

Let us define the following Baxter operators

$$\mathcal{O} = \left(\Lambda_4^{[1]} - D^2\right)\left(\Lambda_3^{[-1]} - D^2\right)\left(\Lambda_2^{[-3]} - D^2\right)\left(\Lambda_1^{[-5]} - D^2\right) , \tag{A.2}$$ {?}

$$\mathcal{O}_D = \left(\Lambda_4^{[1]} - D^2\right)\left(\Lambda_3^{[-1]} - D^2\right)\left(\Lambda_2^{[-3]} - D^2\right)\left(\Lambda_1^{[-5]} - D^2\right) . \tag{A.3}$$ {?}

By direct calculation, we find

$$\mathcal{O} Q_a^{[-4]} = 0 \quad Q^a\, \mathcal{O} = 0 \tag{A.4}$$ {?}

where we can deduce

$$\Lambda_1 = \frac{Q_1^{[3]}}{Q_1^{[1]}} , \quad \Lambda_2 = \frac{Q_{12}^{[2]}}{Q_{12}} \frac{Q_1^-}{Q_1^+} , \quad \Lambda_3 = \frac{Q_{123}^+}{Q_{123}^-} \frac{Q_{12}^{[-2]}}{Q_{12}} , \quad \Lambda_4 = \frac{Q_{123}^{[-3]}}{Q_{123}^{[-1]}} , \tag{A.5}$$ {?}

This gives

$$\begin{aligned} T_{\mathbf{4}} &= \Lambda_1 + \Lambda_2 + \Lambda_3 + \Lambda_4 , \\ T_{\mathbf{6}} &= \Lambda_4^+ \Lambda_3^- + \Lambda_4^+ \Lambda_2^- + \Lambda_4^+ \Lambda_1^- + \Lambda_3^+ \Lambda_2^- + \Lambda_3^+ \Lambda_1^- + \Lambda_2^+ \Lambda_1^- \\ T_{\mathbf{6}} &= \Lambda_4^{[2]} \Lambda_3 \Lambda_2^{[-2]} + \Lambda_4^{[2]} \Lambda_3 \Lambda_1^{[-2]} + \Lambda_4^{[2]} \Lambda_2 \Lambda_1^{[-2]} + \Lambda_3^{[2]} \Lambda_2 \Lambda_1^{[-2]}, \end{aligned} \tag{A.6}$$ {?}

We now write

$$Q_a = \sigma_1\, \hat{q}_a , \quad Q_{ab} = \sigma_2\, \hat{q}_{ab} , \quad Q_{abc} = \sigma_3 \hat{q}_{abc} \tag{A.7}$$ {?}

$$\frac{\sigma_2}{\sigma_1^+ \sigma_1^-} = P_{\mathbf{4}} , \quad \frac{\sigma_1 \sigma_3}{\sigma_2^+ \sigma_2^-} = P_{\mathbf{6}} , \quad \frac{\sigma_2}{\sigma_3^+ \sigma_3^-} = P_{\bar{\mathbf{4}}} , \tag{A.8}$$ {?}

We will see

$$\Lambda_4^{[1]} \Lambda_2^- \Lambda_3^{[-3]} \Lambda_1^{[-5]} = 1 \tag{A.9}$$ {?}

This finally gives the general Baxter equation

$$P_{\mathbf{4}} P_{\mathbf{4}} \tag{A.10}$$ {?}

**The alternating spin chain**

**The isotropic alternating spin chain** Let us finally consider the case $P_{\mathbf{4}} = P_{\bar{\mathbf{4}}} = \mathcal{Q}_\theta$ we find

$$q^{[4]} Q_\theta^{[-3]} \left(Q_\theta^-\right)^2 Q_\theta^+ - \tau_{\mathbf{4}} Q_\theta^{[-3]} Q_\theta^- q^{[2]} + \tau_{\mathbf{6}} Q_\theta^{[-3]} Q_\theta^{[3]} q - \tau_{\bar{\mathbf{4}}} Q_\theta^{[3]} Q_\theta^+ q^{[-2]} + q^{[-4]} Q_\theta^{[3]} \left(Q_\theta^+\right)^2 Q_\theta^- = 0 . \tag{A.11}$$ {?}

| $L$ | States |
|---|---|
| 1 | $[101]^{\otimes 2} = \underbrace{[202]}_{\mathbf{30}} \oplus [210]$ |

**Table 1**: Caption

## B List of States

We list the low-lying states in the alternating spin chain. See table

## References


Ekhammar:2023iph [1] S. Ekhammar, N. Gromov and P. Ryan, *Boundary overlaps from Functional Separation of Variables*, *JHEP* **05** (2024) 268 [2312.11612].